1. Let $f(x)=\begin{cases}\dfrac{x^3+x^2-16x+20}{(x-2)^2} & \text{if } x\neq 2\\ k & \text{if } x=2\end{cases}$. If $f(x)$ is continuous for all x and $k=a^2-b^2$ where $a, b \in N$ then $b=$

माना $f(x)=\begin{cases}\dfrac{x^3+x^2-16x+20}{(x-2)^2} & \text{if } x\neq 2\\ k & \text{if } x=2\end{cases}$. यदि $f(x)$ सभी x के लिए सतत् है तथा $k=a^2-b^2$ जहाँ $a, b \in N$ तो $b=$

(A) 4 (B) 7 (C) 3 (D) none

Ans. C

Sol. Given $f(2)=k$

Now, $\lim\limits_{x\to 2} f(x)=\lim\limits_{x\to 2^+} f(x)$

$$=\lim_{x\to 2}\frac{x^3+x^2-16x+20}{(x-2)^2}$$

$$=\lim_{x\to 2}\frac{(x-5)(x-2)^2}{(x-2)^2}$$

$$=\lim_{x\to 2}(x+5)$$

$=7$

As f is continuous at $x=2$, we have

$\lim\limits_{x\to 2} f(x)=f(2)$

$\Rightarrow k=7$

2. The function $f(x)=\begin{cases}x^2/a & 0\le x<1\\ a & 1\le x<\sqrt{2}\\ (2b^2-4b)/x^2 & \sqrt{2}\le x<\infty\end{cases}$ is continuous for $0\le x<\infty$, then the most suitable values of

a and b are :

(A) $a = 1 , b = -1$ (B) $a = -1 , b = 1 + \sqrt{2}$

(C) $a = -1, b = 1$ (D) none of these

फलन $f(x) = \begin{cases} x^2/a & 0 \le x < 1 \\ a & 1 \le x < \sqrt{2} \\ (2b^2 - 4b)/x^2 & \sqrt{2} \le x < \infty \end{cases}$ अन्तराल $0 \le x < \infty$ में सतत् है तो a तथा b के मान होंगे :

(A) $a = 1 , b = -1$ (B) $a = -1 , b = 1 + \sqrt{2}$

(C) $a = -1, b = 1$ (D) इनमें से कोई नहीं

Ans. C

Sol. Since, f is continuous for $0 \le x < \infty$, f is continuous at $x = 1$.

$$\lim_{x \to 1^-} \frac{x^2}{a} = a$$

$\therefore a = \pm 1$

Since, f is continuous at $x = \sqrt{2}$

$$\lim_{x \to \sqrt{2}} \frac{2b^2 - 4b}{x^2} = a$$

$$\therefore \frac{2b^2 - 4b}{2} = a$$

$\Rightarrow b^2 - 2b = a$

When $a = 1$, $b^2 - 2b = 1$

$\Rightarrow b = 1 \pm \sqrt{2}$

When $a = -1$, $b^2 - 2b = -1$

$\Rightarrow (b - 1)^2 = 0 \Rightarrow b = 1$

Hence, $a = -1$ and $b = 1$ are most suitable values.

3. If $f(x) = \begin{cases} \dfrac{36^x - 9^x - 4^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}} & ; x \ne 0 \\ K & ; x = 0 \end{cases}$ is continuous at $x = 0$ then K equals :

(A) $16\sqrt{2} \cdot \ell n2 . \ell n3$ (B) $16\sqrt{2}\ \ell n(6)$ (C) $16\sqrt{2}\ \ell n(2(\ln 3))$ (D) none of these

यदि $f(x)=\begin{cases}\dfrac{36^x-9^x-4^x+1}{\sqrt{2}-\sqrt{1+\cos x}} & ;\ x\neq 0\\ K & ;\ x=0\end{cases}$ $x=0$ पर सतत् है तो K का मान होगा :

(A) $16\sqrt{2}\,.\ell n2.\ell n3$ (B) $16\sqrt{2}\,\ell n(6)$ (C) $16\sqrt{2}\,\ell n(2(\ln 3))$ (D) इनमें से कोई नहीं

Ans. A

Sol. Function $f(x)$ is continuous at $x=0$

$$\lim_{x\to 0} f(x)=f(0)$$

$$\lim_{x\to 0}\left(\frac{36^x-9^x-4^x+1}{\sqrt{2}-\sqrt{1+\cos x}}\right)=k$$

$$\Rightarrow \lim_{x\to 0}\left(\frac{(9^x-1)(4^x-1)}{\sqrt{2}-\sqrt{1+\cos x}}\right)=k$$

$$\Rightarrow \lim_{x\to 0}\left(\frac{(9^x-1)(4^x-1)}{\sqrt{2}-\sqrt{2}\cos\left(\frac{x}{2}\right)}\right)=k$$

$$\Rightarrow \lim_{x\to 0}\left(\frac{(9^x-1)(4^x-1)}{\sqrt{2}\left\{2\sin^2\left(\frac{x}{4}\right)\right\}}\right)=k$$

$$\Rightarrow \lim_{x\to 0}\left(\frac{8(9^x-1)(4^x-1)}{\sqrt{2}x^2\left\{\frac{2\sin^2\left(\frac{x}{4}\right)}{\frac{x^2}{16}}\right\}}\right)=k$$

$$\frac{8.\ln 9.\ln 4}{\sqrt{2}}=k$$

$$\Rightarrow \frac{32.\ln 3.\ln 2}{\sqrt{2}} = k$$

$$\Rightarrow k = 16\sqrt{2}.\ln 2.\ln 3$$

4. The function $f(x) = \begin{cases} [x] + \sqrt{x-[x]} & \text{for} \quad x \geq 0 \\ \sin x & \text{for} \quad x < 0 \end{cases}$:

(A) cont. only for all non-negative integers
(B) cont. only for all positive integers
(C) discont. only for all negative integers
(D) cont. for all real numbers

फलन $f(x) = \begin{cases} [x] + \sqrt{x-[x]} & \text{for} \quad x \geq 0 \\ \sin x & \text{for} \quad x < 0 \end{cases}$:

(A) केवल अऋणात्मक पूर्णाकों के लिए सतत् होगा
(B) केवल धनात्मक पूर्णाकों के लिए सतत् होगा
(C) केवल ऋणात्मक पूर्णाकों के लिए असतत् होगा
(D) सभी वास्तविक संख्याओं के लिए सतत् है

Ans. D

Sol. $f(x) = \begin{cases} [x] + \sqrt{\{x\}} & \text{for} \quad x \geq 0 \\ \sin x & \text{for} \quad x < 0 \end{cases}$

for +ve integers

$$\lim_{x \to I^+} I + 0 = I$$

$$\lim_{x \to I^-} (I-1) + \sqrt{1} = I$$

$f(I) = I$

At $x = 0 \Rightarrow f(0) = 0$

$$\lim_{x \to 0^-} -1 + \sqrt{1} = 0$$

$$\lim_{x \to 0^+} \sin x = 0$$

Continuous for all real numbers.

5. $y = f(x)$ is a continuous function such that its graph passes through $(a, 0)$. Then $\underset{x \to a}{\text{Limit}} \frac{\log_e (1 + 3f(x))}{2f(x)}$ is :

सतत् फलन $y = f(x)$ इस प्रकार है कि फलन का आरेख $(a, 0)$ से गुजरता है तो $\underset{x \to a}{\text{Limit}} \frac{\log_e (1 + 3f(x))}{2f(x)}$ का मान होगा :

(A) 1 (B) 0 (C) 3/2 (D) 2/3

Ans. C

Sol. $y = f(x)$, passes $(a, 0)$

$\therefore f(a) = 0$

$$= \lim_{x \to a} \frac{\ell n\left(1+3f\left(x\right)\right)}{2f\left(x\right)}$$

$$= \lim_{x \to a} \frac{\ell n\left(1+3f\left(x\right)\right)}{3f\left(x\right)} \frac{3}{2} = \frac{3}{2}$$

6. $f(x) = [2x + a].\ \sin\{x\}$ is continuous at $x = -1$ [where $[\times]$ and $\{\times\}$ respectively denote greatest integer and the fractional part function] if *:*

(A) $a \in [2, 3)$ (B) $a \in (2, 3]$ (C) $a \in [0, 1)$ (D) none of these

यदि $f(x) = [2x + a].\ \sin\{x\}$ $x = -1$ पर सतत है [जहाँ $[\times]$ तथा $\{\times\}$ क्रमशः महत्तम पूर्णांक तथा भिन्नात्मक भाग फलन दर्शाता है] तो :

(A) $a \in [2, 3)$ (B) $a \in (2, 3]$ (C) $a \in [0, 1)$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\text{LHL} = \lim_{h \to 0} f\left(-1-h\right)$

$$= \lim_{h \to 0}\left[-2-h+a\right]\cdot\sin\left\{-1-h\right\}$$

$$= \lim_{h \to 0}\left(-2+\left[a-h\right]\right)\sin\left\{-h\right\}$$

$= (-2 + [a^-])\ \sin 1$

$\text{RHL} = \lim_{h \to 0} f\left(-1+h\right)$

$$= \lim_{h \to 0}\left[-2+a+h\right]\sin\left\{-1+h\right\}$$

$$= \lim_{h \to 0}\left(-2+\left[a+h\right]\right)\sin\left\{h\right\} = 0$$

$\text{LHL} = \text{RHL}$

$-2 + [a^-] = 0$

$\Rightarrow [a^-] = -2$

$\Rightarrow a \in (2, 3]$

7. If $f(x)=\begin{cases} a+\dfrac{\sin[x]}{x} & , \quad x>0 \\ 2 & , \quad x=0 \\ b+\left[\dfrac{\sin x-x}{x^3}\right] & , \quad x<0 \end{cases}$ (where [.] denotes the greatest integer function). If f(x) is continuous at $x=0$, then b is equal to :

यदि $f(x)=\begin{cases} a+\dfrac{\sin[x]}{x} & , \quad x>0 \\ 2 & , \quad x=0 \\ b+\left[\dfrac{\sin x-x}{x^3}\right] & , \quad x<0 \end{cases}$ (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है), $x=0$ पर सतत है तो b का मान होगा :

(A) $a-1$ (B) $a+1$ (C) $a+2$ (D) $a-2$

Ans. B

Sol. $\text{LHL}=\lim\limits_{h\to 0} f(-h)$

$=\lim\limits_{h\to 0} b+\left[\dfrac{\sin(-h)+h}{-h^3}\right]$

$=\lim\limits_{h\to 0} b+\left[\dfrac{h-\sin h}{-h^3}\right]$

$=\lim\limits_{h\to 0} b+\left[\dfrac{h-h+\dfrac{h^3}{3!}-\dfrac{h^5}{5!}.....}{-h^3}\right]$

$=\lim\limits_{h\to 0} b+\left[-\dfrac{1}{3!}+\dfrac{h^2}{5!}.....\right]$

$=\lim\limits_{h\to 0} b-1=b-1$

$\text{RHL}=\lim\limits_{h\to 0} f(h)=\lim\limits_{h\to 0} a+\dfrac{\sin[h]}{h}$

$=\lim\limits_{h\to 0} a+0=a$

LHL = RHL = f(0) $\Rightarrow$ b – 1 = a = 2

b = a + 1

8. Given $f(x) = b([x]^2 + [x]) + 1$ for $x \geq -1$

$= \text{Sin}(\pi(x+a))$ for $x < -1$

where [x] denotes the integral part of x, then for what values of a, b the function is continuous at $x = -1$:

दिया गया है $f(x) = b([x]^2 + [x]) + 1$ for $x \geq -1$

$= \text{Sin}(\pi(x+a))$ for $x < -1$

(जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) तो a, b के मान होंगे यदि फलन $x = -1$ पर सतत है :

(A) $a = 2n + (3/2); b \in R; n \in I$ (B) $a = 4n + 2; b \in R; n \in I$

(C) $a = 4n + (3/2); b \in R^+; n \in I$ (D) $a = 4n + 1; b \in R^+; n \in I$

Ans. A

Sol. $\lim\limits_{x \to -1^+} f(x) = \lim\limits_{x \to -1^+} b([x]^2 + [x]) + 1$

$= \lim\limits_{h \to 0} b([-1 + h]^2 + [-1 + h]) + 1$

$= b((-1)^2 - 1) + 1 = 1$

$\Rightarrow b \in R$

$\lim\limits_{x \to -1^-} f(x) = \lim\limits_{x \to -1^-} \sin(\pi(x + a))$

$= \lim\limits_{h \to 0} \sin(\pi(-1 - h + a)) = -\sin \pi a$

$\sin \pi a = -1$

$\pi a = 2n\pi + \dfrac{3\pi}{2} \Rightarrow a = 2n + \dfrac{3}{2}$

9. The value of f(0) so that the function $f(x) = \dfrac{1}{x} - \dfrac{2}{\left(e^{2x} - 1\right)}$ $(x \neq 0)$ is continuous at x = 0 is:

(A) 0 (B) 1 (C) 2 (D) none of these

f(0) का मान होगा यदि फलन $f(x) = \dfrac{1}{x} - \dfrac{2}{\left(e^{2x} - 1\right)}$ $(x \neq 0)$, x = 0 पर सतत है :

(A) 0 (B) 1 (C) 2 (D) इनमें से कोई नहीं

Ans. B

Sol. $f(x) = \dfrac{1}{x} - \dfrac{2}{e^{2x} - 1}$ can be continuous at x = 0

so $\lim\limits_{h \to 0} f(0 + h) = \lim\limits_{h \to 0} f(0 - h) = f(0)$

$\lim\limits_{h \to 0} f(0 + h) =$

$$\lim_{h\to 0}\frac{1}{h}-\frac{2}{e^{2h}-1}$$

$$\lim_{h\to 0}\frac{(e^{2h}-1)-2h}{h\times(e^{2h}-1)}=\frac{0}{0}\text{ form}$$

$$\lim_{h\to 0}\frac{e^{2h}\times 2-0-2}{h\times e^{2h}\times 2+e^{2h}-1}=\frac{0}{0}\text{ form}$$

$$\lim_{h\to 0}\frac{2\times 2e^{2h}}{2e^{2h}+h\times e^{2h}\times 2\times 2+e^{2h}\times 2}=\frac{4}{4}=1$$

$f(0)=1$

10. Let $f(x)=\dfrac{\log(1+x+x^2)+\log(1-x+x^2)}{\sec x-\cos x}$, $x\neq 0$. The value of $f(0)$ so that f is continuous at $x=0$ is :

(A) 1 (B) 0 (C) 2 (D) none of these

माना फलन $f(x)=\dfrac{\log(1+x+x^2)+\log(1-x+x^2)}{\sec x-\cos x}$, $x\neq 0$ तो $f(0)$ का मान होगा यदि फलन $f\ x=0$ पर सत्त है :

(A) 1 (B) 0 (C) 2 (D) इनमें से कोई नहीं

Ans. A

Sol. $$\lim_{x\to 0}\frac{\log\left(1+x+x^2\right)+\log\left(1-x+x^2\right)}{\sec x-\cos x}$$

$$=\lim_{x\to 0}\frac{\log\left[\left(1+x^2\right)^2-x^2\right]}{\frac{\left(1-\cos^2 x\right)}{\cos x}}$$

$$=\lim_{x\to 0}\frac{\log\left(1+x^2+x^4\right)}{\sin x\tan x}$$

$$=\lim_{x\to 0}\frac{\log\left(1+x^2\left(1+x^2\right)\right)}{x^2\left(1+x^2\right)}\cdot x^2\left(1+x^2\right)\cdot\frac{1}{\frac{\sin x}{x}\cdot\frac{\tan x}{x}\cdot x^2}=1$$

11. Let f(x) be a function defined on [0, 1] such that $f(x)=\begin{cases} x & ; x \in Q \\ 1-x & ; x \notin Q \end{cases}$, then fof(x) is :

(A) continuous for all $x \in [0, 1]$ (B) continuous for only one value of x

(C) not possible to define (D) none of these

माना फलन f(x) अन्तराल [0, 1] में परिभाषित है इस प्रकार कि $f(x)=\begin{cases} x & ; x \in Q \\ 1-x & ; x \notin Q \end{cases}$, तो fof(x) होगा :

(A) सभी $x \in [0, 1]$ के लिए सत्त है (B) x के केवल एक मान के लिए सत्त

(C) परिभाषित करना सम्भव नहीं है (D) इनमें से कोई नहीं

Ans. A

Sol. $f(x)=\begin{cases} f(x) & ; f(x) \in Q \\ 1-f(x) & ; f(x) \notin Q \end{cases}$

$\Rightarrow f(f(x))=\begin{cases} x & ; x \in Q \\ 1-(1-x) & ; x \notin Q \end{cases}$

$\Rightarrow f(f(x))=\begin{cases} x & ; x \in Q \\ x & ; x \notin Q \end{cases}$

$\Rightarrow fof(x)=x, x \in R$

12. Let $f(x)=\frac{1-\tan x}{4x-\pi}, x \neq \frac{\pi}{4}, x \in \left[0, \frac{\pi}{2}\right]$. If f(x) is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is :

मानाकि $f(x)=\frac{1-\tan x}{4x-\pi}, x \neq \frac{\pi}{4}, x \in \left[0, \frac{\pi}{2}\right]$ यदि f(x) अंतराल $\left[0, \frac{\pi}{2}\right]$ में सतत हो , तो $f\left(\frac{\pi}{4}\right)$ का मान है :

(A) 1 (B) 1/2 (C) – 1/2 (D) –1

Ans. C

Sol. $f(x)=\frac{1-\tan x}{4x-\pi}$ $x \neq \pi/4$ $x \in [0, \pi/2]$

f(x) is continuous at $x \in [0, \pi/2]$

so at $x = \pi/4$

$\lim_{h\to 0} f\left(\frac{\pi}{4}+h\right)=\lim_{h\to 0} f\left(\frac{\pi}{4}-h\right)=f\left(\frac{\pi}{4}\right)$

so $\lim_{h\to 0} f\left(\frac{\pi}{4}+h\right)=f\left(\frac{\pi}{4}\right)$

$$\lim_{h\to 0} f\left(\frac{\pi}{4}+h\right)$$

$$= \lim_{h\to 0} \frac{1-\tan\left(\frac{\pi}{4}+h\right)}{4\left(\frac{\pi}{4}+h\right)-\pi}$$

$$= \lim_{h\to 0} \frac{1-\left(\frac{\tan\frac{\pi}{4}+\tanh}{1-\tan\frac{\pi}{4}\tanh}\right)}{\pi+4h-\pi}$$

$$= \lim_{h\to 0} \frac{1-\tanh-1-\tanh}{(1-\tanh)\times 4h}$$

$$= \lim_{h\to 0} \frac{-2}{4}\left(\frac{\tanh}{h}\right) = -\frac{1}{2}$$

$$\lim_{h\to 0} f\left(\frac{\pi}{4}+h\right) = -\frac{1}{2} = f\left(\frac{\pi}{4}\right)$$

$$f\left(\frac{\pi}{4}\right) = -\frac{1}{2}$$

13. If $f(x) = \begin{cases} \frac{a\left|x^2-x-2\right|}{2+x-x^2} & \text{for } x<2 \\ b & \text{for } x=2 \\ \frac{x-[x]}{x-2} & \text{for } x>2 \end{cases}$ where [ ] denote the greatest integer function, is continuous at $x=2$ the find

'a' and 'b'.

माना $f(x) = \begin{cases} \frac{a\left|x^2-x-2\right|}{2+x-x^2} & \text{for } x<2 \\ b & \text{for } x=2 \\ \frac{x-[x]}{x-2} & \text{for } x>2 \end{cases}$ जहाँ [ ] महतम पूर्णांक फलन दर्शाता है, $x=2$ पर सतत् है, तो $a =$ _____ तथा

$b =$ _____.

Ans. $a = 1\ ,\ b = 1$

Sol. $x^2 - x - 2 = (x + 1)(x - 2) < 0 \ \forall\ x \in (-1,2)$

$$\text{LHL} = \lim_{x\to 2^-} \frac{a\left|x^2 - x - 2\right|}{2 + x - x^2}$$

$$= \lim_{x\to 2^-} \frac{a - \left(-\left(x^2 - x - 2\right)\right)}{2 + x - x^2} = \lim_{x\to 2^-} a = a$$

$$\text{RHL} = \lim_{x\to 2^+} \frac{x - [x]}{x - 2} = \lim_{x\to 2^+} \frac{x - 2}{x - 2}$$

$$= \lim_{x\to 2^+} 1 = 1$$

$\text{LHL} = \text{RHL} = f(2)$

$a = 1 = b$

14. If $f(x) = \dfrac{A\cos x + Bx\sin x - 5}{x^4}$ $(x \neq 0)$ is continuous at $x = 0$, then find the value of A and B. Also find $f(0)$.

यदि फलन $f(x) = \dfrac{A\cos x + Bx\sin x - 5}{x^4}$ $(x \neq 0)$, $x = 0$ पर सत्त है तो A, B, $f(0)$ का मान ज्ञात कीजिये।

Ans. $A=5,\ B = 5/2,\ f(0) = -\dfrac{5}{24}$

Sol. Given $\lim\limits_{x\to 0}\left(\dfrac{\left(a\cos x + bx\sin x - 5\right)}{x^4}\right)$

As $x \to 0$, $x^4 \to 0$, the limit must be in the form of $\left(\dfrac{0}{0}\right)$

Thus, $\lim\limits_{x\to 0}\left(a\cos x + bx\sin x - 5\right) = 0$

$\Rightarrow a - 5 = 0$

$\Rightarrow a = 5$

Now, limit

$$= \lim_{x\to 0}\left(\frac{a\cos x + bx\sin x - 5}{x^4}\right)$$

$$= \lim_{x\to 0}\left(\frac{5\cos x + bx\sin x - 5}{x^4}\right)$$

$$= \lim_{x\to 0} \frac{5\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-.....\right)+bx\left(x-\frac{x^3}{3!}+....\right)-5}{x^4}$$

$$= \lim_{x\to 0} \frac{\left(b-\frac{5}{2}\right)x^2+\left(\frac{5}{24}-\frac{b}{6}\right)x^4+....}{x^4}$$

It will provide finite limit only when, co-efficient of x^2 is zero.

Thus, $\left(b-\frac{5}{2}\right)=0$

$$\Rightarrow b=\frac{5}{2}$$

Also, the limit value

$$= \lim_{x\to 0} \frac{\left(\frac{5}{24}-\frac{b}{6}\right)x^4+....}{x^4}$$

$$=\left(\frac{5}{24}-\frac{b}{6}\right)$$

$$=\left(\frac{5}{24}-\frac{5}{12}\right)$$

$$=-\frac{5}{24}$$

15. If $f(x)=\frac{\sin 3\pi x + A\sin 5\pi x + B\sin \pi x}{(x-1)^5}$ for $x \neq 1$ is continuous at $x = 1$, find A and B. Also find $f(1)$.

यदि $f(x)=\frac{\sin 3\pi x + A\sin 5\pi x + B\sin \pi x}{(x-1)^5}$ for $x \neq 1$, $x = 1$ पर सतत है तो A, B तथा $f(1)$ ज्ञात कीजिये।

Ans. $A=-\frac{1}{5}$; $B=-2$; $f(1)=\frac{16}{5}\pi^5$

Sol. $LHL = \lim_{h\to 0} f(1-h)$

$$= \lim_{h\to 0} \frac{\sin 3\pi h + A\sin 5\pi h + B\sin \pi h}{-h^5}$$

$$= \lim_{h\to 0} \frac{\left(3\pi h - \frac{(3\pi h)^3}{3!} + \frac{(3\pi h)^5}{5!} .....\right) + A\left(5\pi h - \frac{(5\pi h)^3}{3!} + \frac{(5\pi h)^5}{5!} .....\right) + B\left(\pi h - \frac{(\pi h)^3}{3!} + \frac{(\pi h)^5}{5!} .....\right)}{-h^5}$$

$$= \lim_{h\to 0} -\frac{\pi(3+5A+B)}{h^4} + \frac{\pi^3}{3!}\frac{(27+125A+B)}{h^2} - \frac{\pi^5}{5!}\left(3^5+5^5A+B\right) + (\text{Terms of "h"})$$

$$\left.\begin{matrix} 3+5A+B=0 \\ 27+125A+B=0 \end{matrix}\right\} A = -\frac{1}{5}, B = -2$$

$$\text{limit} = -\frac{\pi^5}{5!}\left(3^5+5^5A+B\right) = \frac{16}{5}\pi^5$$

$$f(1) = \text{LHL} = \frac{16}{5}\pi^5$$

16. Let : $R \to Q$ be a continuous function then prove that $\sum_{r=1}^{n} f(r) = \sum_{r=1}^{n} f(2r+1)$.

माना $f : R \to Q$ एक सत्त फलन है तो सिद्ध कीजिये $\sum_{r=1}^{n} f(r) = \sum_{r=1}^{n} f(2r+1)$

Ans.

Sol. f(x) should be a constant function

17. If $f(x) = \log_{1+x^2}(\cos x)$, $x \neq 0$ and $f(0) = k$ is continuous at $x = 0$ then $k =$ _____.

यदि $f(x) = \log_{1+x^2}(\cos x)$, $x \neq 0$, $f(0) = k$ पर सत्त है तो $k =$ _____

Ans. $-\frac{1}{2}$

Sol. $\lim_{x\to 0} \frac{\log\cos x}{\log(1+x^2)} = \lim_{x\to 0} \frac{\cos x - 1}{x^2} = -\lim_{x\to 0} \frac{(1-\cos x)}{x^2}$

$$= -\frac{1}{2}$$

18. Let $f(x)$ is defined as follows, $f(x)=\begin{cases}(\cos x-\sin x)^{\cos ec\,x} & , \quad -\frac{\pi}{2}<x<0 \\ a & , \quad x=0 \\ \frac{e^{1/x}+e^{2/x}+e^{3/x}}{ae^{2/x}+be^{3/x}} & , \quad 0<x<\frac{\pi}{2}\end{cases}$

If f(x) is continuous at $x=0$, find 'a' and 'b'.

माना f(x) परिभाषित है इस प्रकार कि $f(x)=\begin{cases}(\cos x-\sin x)^{\cos ec\,x} & , \quad -\frac{\pi}{2}<x<0 \\ a & , \quad x=0 \\ \frac{e^{1/x}+e^{2/x}+e^{3/x}}{ae^{2/x}+be^{3/x}} & , \quad 0<x<\frac{\pi}{2}\end{cases}$.

यदि $f(x)$, $x=0$ पर सतत है तो ' a ' तथा ' b ' का मान ज्ञात कीजिये।

Ans. $a=e^{-1}$, $b=e$

Sol. If f(x) is continuous at $x=0$ then

$$\lim_{x\to 0^-} f(x)=\lim_{x\to 0^+} f(x)=f(0)$$

$\because\ f(0)=a$

LHL

$$=\lim_{x\to 0^-}(\cos x-\sin x)^{\cos ecx} \qquad 1^{\infty}\text{ form}$$

$$=e^{\lim\limits_{x\to 0^-}\cos ecx(\cos x-\sin x-1)}$$

$$=e^{\lim\limits_{x\to 0^-}\cos ecx(\cot x-1-\cos ecx)}=e^{-1}=\frac{1}{e}=a$$

RHL

$$=\lim_{x\to 0^+}\frac{e^{\frac{1}{x}}+e^{\frac{2}{x}}+e^{\frac{3}{x}}}{ae^{\frac{2}{x}}+be^{\frac{3}{x}}}$$

$$=\lim_{x\to 0^+}\frac{e^{\frac{3}{x}}\left(e^{-\frac{2}{x}}+e^{-\frac{1}{x}}+1\right)}{e^{\frac{3}{x}}\left(ae^{-\frac{1}{x}}+b\right)}$$

$$=\frac{0+0+1}{a\times 0+b}=\frac{1}{b}$$

$$\therefore \frac{1}{b}=\frac{1}{e}$$

$$\Rightarrow b=e \;\&\; a=\frac{1}{e}$$

19. Determine a and b so that f is continuous at $x=\frac{\pi}{2}$ where $f(x)=\begin{cases}\frac{1-\sin^3 x}{3\cos^2 x} & \text{if} \quad x<\frac{\pi}{2}\\ a & \text{if} \quad x=\frac{\pi}{2}\\ \frac{b(1-\sin x)}{(\pi-2x)^2} & \text{if} \quad x>\frac{\pi}{2}\end{cases}$.

दिया गया फलन $f(x)=\begin{cases}\frac{1-\sin^3 x}{3\cos^2 x} & \text{if} \quad x<\frac{\pi}{2}\\ a & \text{if} \quad x=\frac{\pi}{2}\\ \frac{b(1-\sin x)}{(\pi-2x)^2} & \text{if} \quad x>\frac{\pi}{2}\end{cases}$, $x=\frac{\pi}{2}$ पर सतत् हो, तो a व b का मान ज्ञात कीजिए।

Ans. a = 1/2, b = 4

Sol. $\text{V.F.}\big|_{x=\frac{\pi}{2}}=a$

$$\text{LHL}\big|_{x=\frac{\pi}{2}}=\lim_{x\to\left(\frac{\pi}{2}\right)^-} f(x)$$

$$=\lim_{x\to\left(\frac{\pi}{2}\right)^-}\left(\frac{1-\sin^3 x}{3\cos^2 x}\right);\ \frac{0}{0}\ \text{form}$$

$$\text{put } x=\frac{\pi}{2}-h$$

$$=\lim_{h\to 0}\left(\frac{1-\sin^3\left(\frac{\pi}{2}-h\right)}{3\cos^2\left(\frac{\pi}{2}-h\right)}\right)$$

$$=\left(\lim_{h\to 0}\frac{1-\cos^3 h}{3\sin^2 h}\right)$$

$$=\lim_{h\to 0}\frac{(1-\cosh)\left(1+\cos^2 h+\cosh\right)}{3(1-\cosh)(1+\cosh)}=\frac{1}{2}$$

$$RHL\big|_{x=\frac{\pi}{2}} = \lim_{x\to\frac{\pi}{2}^{+}} \frac{b(1-\sin x)}{(\pi-2x)^2}$$

put $x = \frac{\pi}{2} + h$

$$= \lim_{h\to 0} \frac{b}{4}\frac{(1-\cosh)}{h^2} = \frac{b}{8}$$

since function is continuous

$$VF\big|_{x=\frac{\pi}{2}} = LHL\big|_{x=\frac{\pi}{2}} = RHL\big|_{x=\frac{\pi}{2}}$$

$$a = \frac{1}{2} = \frac{b}{8}$$

$$a = \frac{1}{2}, b = 4$$

1. The function $f(x)=\begin{cases}\frac{1}{4}\left(3x^2+1\right) & -\infty < x \le 1\\ 5-4x & 1 < x < 4\\ 4-x & 4 \le x < \infty\end{cases}$ is :

(A) continuous at $x=1$ & $x=4$
(B) continuous at $x=1$, discontinuous at $x=4$
(C) continuous at $x=4$, discontinuous at $x=1$
(D) discontinuous at $x=1$ and $x=4$

फलन $f(x)=\begin{cases}\frac{1}{4}\left(3x^2+1\right) & -\infty < x \le 1\\ 5-4x & 1 < x < 4\\ 4-x & 4 \le x < \infty\end{cases}$ होगा :

(A) $x=1$ तथा $x=4$ पर सत्त
(B) $x=1$ पर सत्त तथा $x=4$ पर असत्त
(C) $x=4$ पर सत्त तथा $x=1$ पर असत्त
(D) $x=1$ तथा $x=4$ पर असत्त

Ans. B

Sol. $f\left(1^-\right)=\frac{1}{4}\left(3.1^2+1\right)$

$=1$

$f(1^+)=5-4.1$

$f(1^-)=f(1^+)=f(1)$

so $f(x)$ continuous of $x=1$

$f(4^-)=5-4.4$

$=5-16=-11$

$f(4^+)=4-4$

$=0$

$f(4^-)\neq f(4^+)$

so $f(x)$ is discontinuous at $x=4$

2. The value of $f(0)$ so that the function $f(x)=\frac{\sqrt{4-2x+x^2}-\sqrt{4+2x+x^2}}{\sqrt{2+x}-\sqrt{2-x}}$ is continuous at $x=0$ is :

f(0) का मान होगा यदि $f(x) = \dfrac{\sqrt{4 - 2x + x^2} - \sqrt{4 + 2x + x^2}}{\sqrt{2 + x} - \sqrt{2 - x}}$, x = 0 पर सत्त है :

(A) 2 (B) 1 (C) $-\sqrt{2}$ (D) $-2\sqrt{2}$

Ans. C

Sol. $\lim\limits_{x\to 0} \dfrac{\left(4-2x+x^2\right)-\left(4+2x+x^2\right)}{(2+x)-(2-x)} \times \dfrac{\sqrt{2+x}+\sqrt{2+x}}{\sqrt{4-2x+x^2}+\sqrt{4+2x+x^2}}$

$\lim\limits_{x\to 0} \dfrac{-4x}{2x} \times \dfrac{\sqrt{2+x}+\sqrt{2-x}}{\sqrt{4-2x+x^2}+\sqrt{4+2x+x^2}} = -\dfrac{2\times 2\sqrt{2}}{2+2}$

$-\sqrt{2}$

3. In [1, 3], the function $[x^2 + 1]$, [x] denoting the greatest integer function, is continuous :

(A) for all x (B) for all x except at nine points

(C) for all x except at seven points (D) for all x except at eight points

अन्तराल [1, 3], में फलन $[x^2 + 1]$, सत्त होगा (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) :

(A) सभी x के लिए (B) नौ बिन्दुओं के अतिरिक्त सभी x के लिए

(C) सात बिन्दुओं के अतिरिक्त सभी x के लिए (D) आठ बिन्दुओं के अतिरिक्त सभी x के लिए

Ans. D

Sol. $[x^2 + 1]$ is discontinuous whenever $(x^2 + 1)$ is an integer.

$[x^2 + 1] = [x^2] + 1$

for $x \in [1,3]$

$\therefore x^2 \in (1,9]$

$\therefore x^2 = 1,2,3,4,5,6,7,8,9$ (Discontinuity)

$\therefore x = 1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, 3$

Hence discontinuous at nine values.

4. If $f(x) = \begin{cases} px^2 - q & , \ x \in [0,1) \\ x + 1 & , \ x \in (1,2] \end{cases}$ and f (1) = 2 , then the value of the pair (p , q) for which f (x) cannot be continuous at x = 1 is :

यदि $f(x)=\begin{cases} px^2-q & , \ x\in[0,1) \\ x+1 & , \ x\in(1,2] \end{cases}$ तथा $f(1)=2$ तो युग्म (p, q) का मान ज्ञात कीजिये जिसके लिये फलन $f(x)$, $x=1$ पर असत्त हो :

(A) (2, 0) (B) (1, – 1) (C) (4, 2) (D) (1, 1)

Ans. D

Sol. f(x) is continuous at x = 1 if

$$\lim_{h\to0} f(1+h)=\lim_{h\to0}(1-h)=f(1)=2$$

$$\lim_{h\to0^+} f(1+h)=\lim_{h\to0^+}(1+h+1)=2$$

$$\lim_{h\to0^-} f\left(p(1-h)^2-q\right)=p-q$$

$\therefore$ f(x) is not continuous at $x=1$ if $p-q\neq2$

Hence option (D) is true.

5. If f is a continuous function such that $f\left(\frac{1}{2}\right)=2$, then $\lim\limits_{x\to0} f\left(\frac{1-\cos x}{x(e^x-1)}\right)$ is :

यदि f एक सत्त फलन है जो इस प्रकार है कि $f\left(\frac{1}{2}\right)=2$, तो $\lim\limits_{x\to0} f\left(\frac{1-\cos x}{x(e^x-1)}\right)$ का मान होगा :

(A) 0 (B) 1/2 (C) 2 (D) None of these

Ans C

6. Given :

$$f(x)=\begin{cases} x & \text{for } |x|\le1 \\ 1 & \text{for } |x|>1 \end{cases}, g(x)=\begin{cases} \cos\left(\frac{\pi}{2}x\right) & \text{for } |x|\le1 \\ |x-1| & \text{for } |x|>1 \end{cases}, h(x)=\begin{cases} \frac{(|x|-1)}{\log_a|x|} & \text{for } |x|\neq1 \\ \ell n\, a & \text{for } |x|=1,\ a>0,\ a\neq1 \end{cases}$$

If ℓ, m, n denote the number of points of discontinuity of the functions f, g and h in their domains respectively, then (ℓ, m, n) is :

दिया गया है :

$$f(x)=\begin{cases} x & \text{for } |x|\le1 \\ 1 & \text{for } |x|>1 \end{cases}, g(x)=\begin{cases} \cos\left(\frac{\pi}{2}x\right) & \text{for } |x|\le1 \\ |x-1| & \text{for } |x|>1 \end{cases}, h(x)=\begin{cases} \frac{(|x|-1)}{\log_a|x|} & \text{for } |x|\neq1 \\ \ell n\, a & \text{for } |x|=1,\ a>0,\ a\neq1 \end{cases}$$

यदि ℓ, m, n फलन f, g तथा h के प्रान्त में स्थित असत्ता के बिन्दुओं की संख्या निरूपित करते हैं तो (ℓ, m, n) होगा :

(A) (0, 0, 0) (B) (1, 1, 1) (C) (2, 2, 2) (D) (1, 1, 0)

Ans. D

Sol. Discontinuity at x = 1

$$\lim_{x\to 1} h(x) \Rightarrow \lim_{x\to 1} \frac{|x|-1}{\frac{\log|x|}{\log a}}$$

$$= \lim_{x\to 1} \log a \frac{(|x|-1)}{(|x|-1)} = \log a$$

h(x) is continuous everywhere.

7. Let $f(x)=\begin{cases} x^2-4x+3 & x<3 \\ x-4 & x\geq 3 \end{cases}$ and $g(x)=\begin{cases} x-3 & x\geq 4 \\ x^2-x & 1<x<4 \\ 1 & x=1 \end{cases}$ then function f(x) × g(x) is discontinuous at :

(A) exactly 1 point (B) exactly 2 points (C) exactly 3 points (D) None

माना $f(x)=\begin{cases} x^2-4x+3 & x<3 \\ x-4 & x\geq 3 \end{cases}$ तथा $g(x)=\begin{cases} x-3 & x\geq 4 \\ x^2-x & 1<x<4 \\ 1 & x=1 \end{cases}$ तो फलन f(x) × g(x) असत्त होगा :

(A) ठीक 1 बिन्दु पर (B) ठीक 2 बिन्दु पर (C) ठीक 3 बिन्दु पर (D) कोई नहीं

Ans. A

Sol. Let $h(x)=f(x)\times g(x)=\begin{cases} 0 & x=1 \\ (x^2-4x+3)(x^2-x) & 1<x<3 \\ (x-4)(x^2-x) & 3\leq x<4 \\ (x-4)(x-3) & x\geq 4 \end{cases}$

at x = 1

$$RHL = \lim_{x\to 1^+} f(x)g(x) = \lim_{x\to 1^+} (x^2-4x+3)(x^2-x)$$

= 0 = h(1)

at x = 3

$$LHL = \lim_{x\to 3^-} h(x) = \lim_{x\to 3^-} (x^2-4x+3)(x^2-x) = 0$$

$$RHL = \lim_{x\to 3^+} h(x) = \lim_{x\to 3^+} (x-4)(x^2-x) = -6$$

Discontinuous at x = 3

at $x = 4$

$$LHL = \lim_{x \to 4^-} h(x) = \lim_{x \to 4^-} (x-4)(x^2 - x) = 0$$

$$RHL = \lim_{x \to 4^+} h(x) = \lim_{x \to 4^+} (x-4)(x-3) = 0$$

$LHL = RHL = f(4)$

8. The function $f(x) = \dfrac{4 - x^2}{|4x - x^3|}$ is (in domain) :

(A) discontinuous at only one point
(B) discontinuous at exactly two points
(C) discont. at exactly three points
(D) none

फलन $f(x) = \dfrac{4 - x^2}{|4x - x^3|}$ असत्त होगा (प्रान्त में) :

(A) ठीक 1 बिन्दु पर
(B) ठीक 2 बिन्दु पर
(C) ठीक 3 बिन्दु पर
(D) कोई नहीं

Ans. D

Sol. $f(x) = \dfrac{4 - x^2}{|x||4 - x^2|}$

$$f(x) = \begin{cases} \dfrac{1}{|x|} & |x < 2 - \{0\}| \\ -\dfrac{1}{|x|} & |x| > 2 \end{cases}$$

Discontinuous at $x = 0, 2, -2$

But $\{0, 2, -2\} \notin D_f$

function is continuous at every point in domain.

9. The jump of discontinuity for the function, (where [x] denotes greastest integer function and

$\{x\}$ denotes fractional part of x, $f(x) = \begin{cases} \frac{e^{[x]} - e^{\{x\}}}{e^x} & , \quad x < 0 \\ \frac{\sin\{x\}}{\{\tan x\}} & , \quad x > 0 \\ 2 & , \quad x = 0 \end{cases}$:

फलन f के लिए असत्ता का jump होगा (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है तथा $\{x\}$ भिन्नात्मक भाग फलन दर्शाता है)

$f(x) = \begin{cases} \frac{e^{[x]} - e^{\{x\}}}{e^x} & , \quad x < 0 \\ \frac{\sin\{x\}}{\{\tan x\}} & , \quad x > 0 \\ 2 & , \quad x = 0 \end{cases}$:

(A) $e^{-1} - e - 1$ (B) $e^{-1} - e - 2$ (C) 1 (D) $1 + e - e^{-1}$

Ans. D

Sol. $LHL = \lim_{x \to 0^-} \frac{e^{[x]} - e^{\{x\}}}{e^x} = \frac{e^{-1} - e}{1}$

$= e^{-1} - e$

$RHL = \lim_{x \to 0^+} \frac{\sin\{x\}}{\{\tan x\}} = \lim_{x \to 0^+} \frac{\{x\}}{\{x\}} = 1$

$JUMP = |RHL - LHL|$

$= 1 - e^{-1} + e = 1 + e - e^{-1}$

10. Consider the function defined on $[0, 1] \to R$, $f(x) = \frac{\sin x - x \cos x}{x^2}$ if $x \neq 0$ and $f(0) = 0$, then the function $f(x)$:

(A) has a removable discontinuity at $x = 0$

(B) has a non removable finite discontinuity at $x = 0$

(C) has a non removable infinite discontinuity at $x = 0$

(D) is continuous at $x = 0$

यदि $f : [0, 1] \to R$ तथा $f(x) = \frac{\sin x - x \cos x}{x^2}$, $x \neq 0$ और $f(0) = 0$ तो फलन $f(x)$:

(A) $x = 0$ पर विस्थापनीय असतत्ा

(B) $x = 0$ पर अविस्थापनीय परिमित असतत्ा

(C) $x = 0$ पर अविस्थापनीय तथा अपरिमित (अनन्त) असतत्ा

(D) $x = 0$ पर फलन सतत् है

Ans. D

Sol. $\lim\limits_{x\to0} f(x) = \lim\limits_{x\to0} \dfrac{\left(x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!}.....\right) - x\left(1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!}.....\right)}{x^2}$

$= \lim\limits_{x\to0}\left(\dfrac{1}{2!} - \dfrac{1}{3!}\right)x + \left(\dfrac{1}{5!} - \dfrac{1}{4!}\right)x^3.....$

$= 0$

$\lim\limits_{x\to0} f(x) = f(0)$

Hence continuous at $x = 0$.

11. Examine the continuity of the function defined by

$f(x) = \dfrac{|x-2|}{x-2}, \quad x \neq 2$

$= 1, \quad x = 2$ at point $x = 2$:

फलन की $x = 2$ पर सांतत्यता की जाँच कीजिये

$f(x) = \dfrac{|x-2|}{x-2}, \quad x \neq 2$

$= 1, \quad x = 2$ at point $x = 2$:

Ans.

Sol. $f(x) = \begin{cases} -1 & x < 2 \\ 1 & x \geq 2 \end{cases}$

at $x = 2$ function has Jump discontinuity

12. If $f(x) = \begin{cases} a\sin x - \cos x, & 0 < x < \pi/2 \\ 2b\cos 2x + \sin x, & \dfrac{\pi}{2} \leq x \leq \pi \\ (a+1)\tan\left(\dfrac{\pi}{4} + x\right), & \pi < x < \dfrac{3\pi}{4} \end{cases}$ is continuous at $x = \dfrac{\pi}{2}$ and $x = \pi$, the find 'a' and 'b'.

यदि $f(x) = \begin{cases} a\sin x - \cos x, & 0 < x < \pi/2 \\ 2b\cos 2x + \sin x, & \dfrac{\pi}{2} \leq x \leq \pi \\ (a+1)\tan\left(\dfrac{\pi}{4} + x\right), & \pi < x < \dfrac{3\pi}{4} \end{cases}$, $x = \dfrac{\pi}{2}$ तथा $x = \pi$ पर सतत है तो 'a' तथा 'b' के मान ज्ञात कीजिये।

Ans. $a = 0, b = \frac{1}{2}$

Sol. At $x = \frac{\pi}{2}$

$$LHL = \lim_{x \to \frac{\pi}{2}^-} a \sin x - \cos x = a$$

$$RHL = \lim_{x \to \frac{\pi}{2}^+} 2b \cos 2x + \sin x = -2b + 1$$

$LHL = RHL \Rightarrow a = -2b + 1$ ...(1)

at $x = \pi$

$$LHL = \lim_{x \to \pi^-} 2b \cos 2x + \sin x = 2b$$

$$RHL = \lim_{x \to \pi^+} (a+1) \tan\left(\frac{\pi}{4} + x\right) = a + 1$$

$LHL = RHL \Rightarrow 2b = a + 1$ ...(2)

from (1) & (2)

$a = 0, \ b = \frac{1}{2}$

13. Examine the continuity of the function defined by $f(x) = \begin{cases} -x^2, x \le 0 \\ 5x - 4, 0 < x \le 1 \\ 4x^2 - 3x, 1 < x < 2 \\ 3x + 4, x \ge 2 \end{cases}$ at the points $x = 0, 1, 2$.

फलन $f(x) = \begin{cases} -x^2, x \le 0 \\ 5x - 4, 0 < x \le 1 \\ 4x^2 - 3x, 1 < x < 2 \\ 3x + 4, x \ge 2 \end{cases}$ की $x = 0, 1, 2$ पर सांतत्यता की जाँच कीजिये।

Ans. disc at $x = 0$, cont. at $x = 1, 2$

Sol. at $x = 0$

$$LHL = \lim_{x \to 0^-} -x^2 = 0$$

$$RHL = \lim_{x \to 0^+} (5x - 4) = -4$$

at $x = 0$, f(x) has Jump discontinuity

at x = 1

$LHL = \lim_{x \to 1^-} 5x - 4 = 1$

$RHL = \lim_{x \to 1^+} 4x^2 - 3x = 1$

LHL = RHL = f(1)

f(x) is continuous at x = 1

at x= 2 $\Rightarrow LHL = \lim_{x \to 2^-} 4x^2 - 3x = 10$

$\Rightarrow RHL = \lim_{x \to 2^+} 3x + 4 = 10$

LHL = RHL = f(2), f(x) is continuous at x = 2.

14. Examine the continuity of the function $f(x) = \begin{cases} 2x - [x] + \sin\frac{1}{x}, & x \neq 0 \\ 0 \quad , & x = 0 \end{cases}$. where [x] denote the greatest integer not greater than x; at x = 0 and x = 2.

फलन f की x = 0 तथा x = 2 पर सांतत्यता की जाँच कीजिये। $f(x) = \begin{cases} 2x - [x] + \sin\frac{1}{x}, & x \neq 0 \\ 0 \quad , & x = 0 \end{cases}$ (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

Ans. discontinuous at x = 0, 2

Sol. $f(x) = \begin{cases} x + \{x\} + \sin\frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$

f(x) is discontinuous $\forall\ x \in I$.

15. A function f(x) is defined by

$f(x) = \begin{cases} \frac{[x^2]-1}{x^2-1} & \text{for } x^2 \neq 1 \\ 0 & \text{for } x^2 = 1 \end{cases}$, where [.] represent greatest integer function.

Discuss the continuity of f(x) at x = 1.

फलन f की x = 1 पर सांतत्यता की जाँच कीजिये।

$f(x) = \begin{cases} \frac{[x^2]-1}{x^2-1} & \text{for } x^2 \neq 1 \\ 0 & \text{for } x^2 = 1 \end{cases}$, (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

Ans. discontinuous at x = 1

Sol. $LHL = \lim_{x\to 1^-} \frac{[x^2]-1}{x^2-1} = \lim_{x\to 1^-} \frac{-1}{x^2-1} = ND$

$RHL = \lim_{x\to 1^+} \frac{[x^2]-1}{x^2-1} = 0$

$LHL \neq RHL$, Discontinuous at $x = 1$

16. Let $f(x) = \begin{cases} (\cos 2x)^{\cot^2 2x} & x > 0 \\ p & x = 0 \\ \left(\frac{\sin x}{x}\right)^{3/x^2} & x < 0 \end{cases}$ then find the possible value of p such that f(x) may be continuous at $x = 0$.

माना $f(x) = \begin{cases} (\cos 2x)^{\cot^2 2x} & x > 0 \\ p & x = 0 \\ \left(\frac{\sin x}{x}\right)^{3/x^2} & x < 0 \end{cases}$ तो p के सम्भव मान ज्ञात कीजिये यदि f(x), $x = 0$ पर सत्त हो।

Ans. $p = \frac{1}{\sqrt{e}}$

Sol. $LHL = \lim_{x\to 0^-} (\cos 2x)^{\cot^2 2x}$ (1^∞)

$= e^{\lim_{x\to 0^-}\left(\frac{\cos 2x-1}{\tan^2 2x}\right)}$

$= e^{\lim_{x\to 0^-} -\left(\frac{1-\cos 2x}{(2x)^2}\right)}$

$= e^{-\frac{1}{2}}$

$RHL = \lim_{x\to 0^+} \left(\frac{\sin x}{x}\right)^{\frac{3}{x^2}}$ (1^∞)

$= e^{\lim_{x\to 0^+}\left(\frac{\sin x-x}{x}\right)\times\frac{3}{x^2}}$

$= e^{\lim_{x\to 0^+} \frac{\left\{x-\frac{x^3}{3!}+\frac{x^5}{5!}.....x\right\}}{x^3}\times 3}$

$= e^{\lim\limits_{x\to 0^+} 3\left[-\frac{1}{3!}+\frac{x^2}{5!}.....\right]} = e^{-\frac{1}{2}}$

LHL = RHL = f(0) $\Rightarrow p = e^{-\frac{1}{2}}$

17. Given the function, $f(x) = x\left[\frac{1}{x(1+x)} + \frac{1}{(1+x)(1+2x)} + \frac{1}{(1+2x)(1+3x)} + \text{..... upto } \infty\right]$.

Find f(0) if f(x) is continuous at x = 0.

दिया गया है फलन $f(x) = x\left[\frac{1}{x(1+x)} + \frac{1}{(1+x)(1+2x)} + \frac{1}{(1+2x)(1+3x)} + \text{..... upto } \infty\right]$. f(0) ज्ञात कीजिये यदि f(x), x = 0 पर सतत है।

Ans. $f(x) = \begin{cases} \frac{2}{1+x} & \text{if } x \neq 0 \text{ and } n \to \infty \\ 2 & \text{if } x = 0 \text{ for cont.} \end{cases}$

Sol. $f(x) = \left\{\frac{1}{1+x} + \frac{x}{(1+x)(1+2x)} + \frac{x}{(1+2x)(1+3x)} + \text{.....upto } \infty\right\}$

$f(x) = \left[\frac{1}{1+x} + \frac{1}{1+x} - \frac{1}{1+2x} + \frac{1}{1+2x} - \frac{1}{1+3x}.....\right]$

$f(x) = \frac{2}{1+x}$

$f(x) = \begin{cases} \frac{2}{1+x} & x \neq 0 \\ 2 & x = 0 \end{cases}$

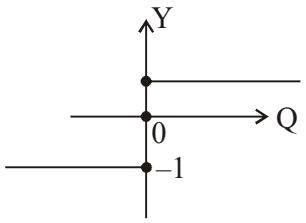
18. The set of all points of discontinuity of fofof where f(x) = sgn(x) is _____.

फलन fofof के असतता के सभी बिन्दु होंगे जहाँ f(x) = sgn(x) _____

Ans. {0}

Sol. f(x) = sgn(x)

$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$



$$f\left(f\left(x\right)\right)=\begin{cases}1 & f\left(x\right)>0 \quad x>0\\ 0 & f\left(x\right)=0 \quad x=0\\ -1 & f\left(x\right)<0 \quad x<-1\end{cases}$$

$$f\left(f\left(f\left(x\right)\right)\right)=\begin{cases}1 & x>0\\ 0 & x=0\\ -1 & x<0\end{cases}$$

So set of all points of continuous is R – {0}

1. Let $f(x)=\begin{cases} 1 & \text{for } x<0 \\ 1+\sin x & 0\le x\le \pi/2 \\ 2+(x-\pi/2)^2 & x\ge \pi/2 \end{cases}$ then which of the following is not correct :

(A) f differentiable at $x=\pi/2$

(B) f is differentiable at $x=0$

(C) f is differetiable at points other than $0, \pi/2$

(D) f is continuous at $x=\pi/2$ and $x=0$

यदि $f(x)=\begin{cases} 1 & \text{for } x<0 \\ 1+\sin x & 0\le x\le \pi/2 \\ 2+(x-\pi/2)^2 & x\ge \pi/2 \end{cases}$ तो निम्न में से कौनसा गलत है :

(A) f, $x=\pi/2$ पर अवकलनीय है

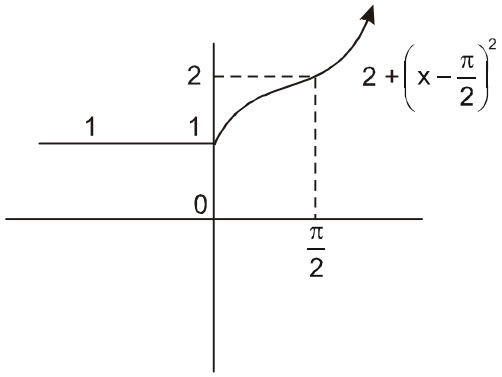
(B) f, $x=0$ पर अवकलनीय है

(C) f, 0 तथा $\pi/2$ के अलावा सभी बिन्दुओं पर अवकलनीय है

(D) f, $x=\pi/2$ तथा $x=0$ पर संत्त है

Ans. B

Sol. $y=f(x)$ is as follows



continous every where सभी स्थानों पर सतत्

non-differentiable at $x=0$ पर अवकलनीय हैं

2. If [.] denotes greatest integer, then function $\cos \pi(|x|+[x])$ is :

(A) not continuous at $x=-1/2$ (B) continuous at $x=0$

(C) differentiable in $[-1, 0)$ (D) differentiable in $(0, 1]$

यदि (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है), फलन $f(x) = \cos \pi (|x| + [x])$ है तो :

(A) $x = -1/2$ संत्त नहीं है

(B) $x = 0$ पर संत्त है

(C) $[-1, 0)$ में अवकलनीय है

(D) $(0, 1]$ में अवकलनीय है

Ans. C

Sol. $\cos \pi\left(|x|+[x]\right)$

(A) at $x = \frac{1}{2}$

$$\text{LHL} = \lim_{h\to 0} \cos \pi\left(\left|-\frac{1}{2}-h\right|+\left[\frac{1}{2}-h\right]\right)$$

$$= \lim_{h\to 0} \cos \pi\left(\frac{1}{2}+h-1\right)$$

$$= \lim_{h\to 0} \cos \pi\left(h-\frac{1}{2}\right) = 0$$

$$\text{RHL} = \lim_{h\to 0} \cos \pi\left(\left|-\frac{1}{2}+h\right|+\left[-\frac{1}{2}+h\right]\right)$$

$$= \lim_{h\to 0} \cos \pi\left(-\left(\frac{1}{2}-h\right)+(-1)\right)$$

$$= \lim_{h\to 0} \cos\left(-\frac{3}{2}+h\right)\pi$$

$$= \lim_{h\to 0} \cos\left(\frac{3\pi}{2}-\pi h\right) = -\lim_{h\to 0} \sin \pi h = 0$$

$$f\left(-\frac{1}{2}\right) = \cos \pi\left(\left|-\frac{1}{2}\right|+\left(-\frac{1}{2}\right)\right) = 0$$

continuous at $x = -\frac{1}{2}$

(B) at $x = 0$

$$\text{LHL} = \lim_{h\to 0} \cos \pi\left(|0-h|+[0-h]\right) = \lim_{h\to 0} \cos(h-1)\pi = -1$$

$$\text{RHL} = \lim_{h\to 0} \cos \pi\left(|0+h|+[0+h]\right) = \lim_{h\to 0} \cos \pi h = 1$$

not continuous at $x = 0$.

(C) $f(x) = \cos\pi(|x| + [x])$

check point $|x| = 0,\ [x] \in I \Rightarrow x = -1,\ x \in [-1,0)$

check point $x = -1$ for $x \in [-1, 0) \Rightarrow [x] = -1$

$\Rightarrow f(x) = \cos\pi(-x-1) \quad \forall x \in [-1,0)$

$f(x) = \cos(\pi(x+1))$

(D) $\forall x \in (0,1]$

$$f(x) = \begin{cases} \cos(\pi(x-0)) & x \in (0,1) \\ \cos\pi(1-1) = 1 & x = 1 \end{cases} = \begin{cases} \cos\pi x & x \in (0,1) \\ 1 & x = 1 \end{cases}$$

$$f'(x) = \begin{cases} -\pi\sin\pi x & x \in (0,1) \\ 0 & x = 1 \end{cases}$$

$$\text{LHD} = f'(1^-) = \lim_{h\to 0} -\pi\sin(\pi(1-n)) = 0$$

Differentiable at $x = 1$.

3. If $f(x) = \begin{cases} \dfrac{x^2-1}{x^2+1} & , \quad 0 < x \le 2 \\ \dfrac{1}{4}(x^3 - x^2) & , \quad 2 < x \le 3 \\ \dfrac{9}{4}(|x-4| + |2-x|) & , \quad 3 < x < 4 \end{cases}$, then :

(A) f(x) is differentiable at $x = 2$ and $x = 3$

(B) f(x) is non-differentiable at $x = 2$ and $x = 3$

(C) f(x) is differentiable at $x = 3$ but not at $x = 2$

(D) f(x) is differentiable at $x = 2$ but not at $x = 3$

यदि $f(x)=\begin{cases} \frac{x^2-1}{x^2+1} & , \quad 0<x\le 2 \\ \frac{1}{4}\left(x^3-x^2\right) & , \quad 2<x\le 3 \\ \frac{9}{4}\left(|x-4|+|2-x|\right) & , \quad 3<x<4 \end{cases}$, तो :

(A) f(x), x = 2 तथा x = 3 पर अवकलनीय है

(B) f(x), x = 2 तथा x = 3 पर अवकलनीय नहीं है

(C) f(x), x = 3 पर अवकलनीय है परन्तु x = 2 पर नहीं है

(D) f(x), x = 2 पर अवकलनीय है परन्तु x = 3 पर नहीं है

Ans. B

Sol. $\lim\limits_{x\to 2^-} f(x)=\frac{3}{5}=f(2)\neq \lim\limits_{x\to 2^+} f(x)=1$

f(x) is not continous at x = 2

$\lim\limits_{x\to 3^-} f(x)=\lim\limits_{x\to 3^+} f(x)=f(3)=\frac{9}{2}$

Now LHD (x =3)

$$\lim_{h\to 0} \frac{\frac{1}{4}((3-h)^3-(3-h)^2)-\frac{9}{2}}{-h} = \lim_{h\to 0} \frac{h^2-8h+21}{4}=\frac{21}{4}$$

and तथा RHD (x = 3) is $\lim\limits_{h\to 0} \frac{\frac{9}{4}(|h-1|+|-1-h|)-\frac{9}{2}}{h}=0$

f(x) is not differentiable at x =2 and x = 3

4. If f(x) is differentiable at x = 4 and f′(4) = 5 then $\lim\limits_{h\to 0} \frac{\left|f(4(1+h^2))\right|-\left|f(4)\right|}{2h^2}$ is equal to :

(A) 10 if and only if $f(4)\neq 0$ (B) 10 if $f(4)>0$

(C) Not exist if f(4) = 0 (D) 10

यदि f(x), x = 4 पर अवकलनीय है तथा f′(4) = 5 है तो $\lim\limits_{h\to 0} \frac{\left|f(4(1+h^2))\right|-\left|f(4)\right|}{2h^2}$ का मान होगा :

(A) 10 यदि और केवल यदि $f(4)\neq 0$ (B) 10 यदि $f(4)>0$

(C) विद्यमान नहीं है यदि f(4) = 0 (D) 10

Ans. B

Sol. f(x) = differentiable at x = 4

$f'(4) = 5$, then $\lim_{h\to 0} \frac{\left|f\left(4\left(1+h^2\right)\right)\right| - \left|f(4)\right|}{2h^2}$

$= 2\lim_{h\to 0} \frac{\left|f\left(4+(2h)^2\right)\right| - \left|f(4)\right|}{(2h)^2}$ $\qquad \left|\lim_{h\to 0} \frac{f(4+h)-f(4)}{h} = f'(4) = 5\right.$

$= 2\lim_{h\to 0} \frac{\left|f(4+h)\right| - \left|f(4)\right|}{h}$

(if $f(4) > 0 \Rightarrow f(4+h) > 0$)

$= 2\lim_{h\to 0} \frac{f(4+h)-f(4)}{h} = 2\times 5 = 10$

5. If f is an even function such that $\lim_{h\to 0^-} \frac{f(h)-f(0)}{h}$ has some finite non-zero value , then :

(A) f is continuous and derivable at x = 0 (B) f is continuous but not derivable at x = 0

(C) f is discontinuous at x = 0 (D) none of these

यदि f एक सम फलन इस प्रकार है कि $\lim_{h\to 0^-} \frac{f(h)-f(0)}{h}$ का मान निश्चित (अशून्य) है तो :

(A) f, x = 0 पर संत्त तथा अवकलनीय है (B) f, x = 0 पर संत्त है परन्तु अवकलनीय नहीं है

(C) f, x = 0 पर असंत्त है (D) इनमें से कोई नहीं

Ans. B

Sol. f(x) = f(–x)

$\lim_{h\to 0^-} \frac{f(h)-f(0)}{h} = f'\left(0^-\right) =$ non zero finite

$\text{LHD} = \lim_{h\to 0} \frac{f(-h)-f(0)}{-h} = \lim_{h\to 0} \frac{f(h)-f(0)}{-h} = -\lim_{h\to 0} \frac{f(h)-f(0)}{h}$

$\text{RHD} = \lim_{h\to 0} \frac{f(h)-f(0)}{h}$

LHD ≠ RHD but finite

⇒ Not differentiable and continuous

6. The function $g(x)=\begin{bmatrix} x+b, & x<0 \\ \cos x, & x\geq 0 \end{bmatrix}$ can be made differentiable at $x=0$:

(A) if b is equal to zero
(B) if b is not equal to zero
(C) if b takes any real value
(D) for no value of b

यदि फलन $g(x)=\begin{bmatrix} x+b, & x<0 \\ \cos x, & x\geq 0 \end{bmatrix}$ को $x=0$ पर अवकलनीय किया जा सकता है :

(A) यदि b का मान शून्य है
(B) यदि b का मान शून्य नहीं है
(C) यदि b का कोई भी वास्तविक मान हो सकता है
(D) b का कोई मान नहीं है

Ans. D

Sol. $g(x)=\begin{cases} x+b & x<0 \\ \cos x & x\geq 0 \end{cases}$ differentiable at $x=0$

$$\text{LHD}=\lim_{h\to 0}\frac{g(-h)-g(0)}{-h}=\lim_{h\to 0}\frac{(-h+b)-1}{-h}=\lim_{h\to 0}\frac{(b-1)-h}{-h}=1\text{, if } b=1 \text{ otherwise DNE}$$

$$\text{RHD}=\lim_{h\to 0}\frac{g(h)-g(0)}{h}=\lim_{h\to 0}\frac{\cosh-1}{h}=0$$

$\Rightarrow$ not differentiable

7. If [.] denotes greatest integer , then for the function $\sin\pi(|x|+[x])$ which of the following is incorrect :

(A) not continuous at $x=-1/2$
(B) continuous at $x=0$
(C) differentiable in $[-1,0)$
(D) differentiable in $(0,1]$

यदि (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) , तो फलन $f(x)=\sin\pi(|x|+[x])$ के लिये निम्न में से कौनसा असत्य है :

(A) $x=-1/2$ पर असंत्त है
(B) $x=0$ पर संत्त है
(C) $[-1,0)$ में अवकलनीय है
(D) $(0,1]$ में अवकलनीय है

Ans. A

Sol. $f(x)=\sin\pi(|x|+[x])$

$$\text{LHL}=\lim_{h\to 0} f\left(-\frac{1}{2}-h\right)=\lim_{h\to 0}\sin\pi\left(h+\frac{1}{2}+\left[-\frac{1}{2}-h\right]\right)$$

$$=\lim_{h\to 0}\sin\pi\left(h+\frac{1}{2}-1\right)$$

$$=\lim_{h\to 0}\sin\pi\left(h-\frac{1}{2}\right)=-1$$

$$RHL = \lim_{h\to 0} f\left(-\frac{1}{2}+h\right) = \lim_{h\to 0} \sin\pi\left(+\frac{1}{2}-h+-1\right) = -1$$

$$f\left(-\frac{1}{2}\right) = \sin\pi\left(\frac{1}{2}-1\right) = -1$$

$$LHL = RHL = f\left(-\frac{1}{2}\right) \Rightarrow \text{continuous}$$

At $x = 0$

$$LHL = \lim_{h\to 0} \sin\pi(h-1) = 0$$

$$RHL = \lim_{h\to 0} \sin\pi(h) = 0$$

$f(0) = 0 \Rightarrow$ continuous at $x = 0$

$$f(x) = \{\sin\pi(-x-1) = +\sin\pi x \quad x \in [-1, 0)$$

differentiable

$$f(x) = \sin\pi(x) \qquad x \in (0, 1]$$

Differentiable

8. Let $g(x) = \begin{bmatrix} a\sqrt{x+2}, & 0 < x < 2 \\ bx+2, & 2 \le x < 5 \end{bmatrix}$. If $g(x)$ is derivable on $(0, 5)$, then find $(2a + b)$.

माना $g(x) = \begin{bmatrix} a\sqrt{x+2}, & 0 < x < 2 \\ bx+2, & 2 \le x < 5 \end{bmatrix}$ है, यदि $g(x)$ एक $(0, 5)$ में अवकलनीय फलन है तो $(2a + b)$ का मान बताइयें।

Ans. 3

Sol. As $g(x)$ is continuous at $x = 2$, so

$g(2) = 2b + 2 = g(2^+)$

and $g(2^-) = \underset{h\to 0}{\text{Lim}}\, a\sqrt{4-h} = 2a$

$\therefore \quad 2a = 2b + 2 \Rightarrow a = b + 1$ .....(1)

$$\text{Now } g'(2^-) = \underset{h\to 0}{\text{Lim}} \frac{f(2-h)-f(2)}{-h} = \underset{h\to 0}{\text{Lim}} \frac{a\sqrt{4-h}-(2b+2)}{-h} = \underset{h\to 0}{\text{Lim}} \frac{a\sqrt{4-h}-(2(a-1)+2)}{-h}$$

$$= \underset{h\to 0}{\text{Lim}} \frac{a\sqrt{4-h}-2a}{-h} = a \underset{h\to 0}{\text{Lim}} \frac{-h}{-h\left(\sqrt{4-h}+2\right)} = \frac{a}{4}$$

$$g'(2^+) = \lim_{h\to 0} \frac{b(2+h)+2-(2b+2)}{h} = \lim_{h\to 0} \frac{bh}{h} = b$$

$\Rightarrow$ $a = 4b$. .....(2)

$\therefore$ On solving (1) and (2), we get $a = \frac{4}{3}$, $b = \frac{1}{3}$

Hence $(2a + b) = 2\left(\frac{4}{3}\right) + \frac{1}{3} = \frac{8}{3} + \frac{1}{3} = 3$

9. The function $f(x) = (x^2 - 1)\,|x^2 - 3x + 2| + \cos(|x|)$ is NOT differentiable at :

फलन $f(x) = (x^2 - 1)\,|x^2 - 3x + 2| + \cos(|x|)$ कहाँ पर अवकलनीय नहीं है :

(A) -1 (B) 0 (C) 1 (D) 2

Ans. D

Sol. $f(x) = (x^2-1)|x^2-3x+2| + \cos(|x|)$

$f(x) = (x^2-1)|x-2||x-1| + \cos(|x|)$

$$f(x) = \begin{cases} (x^2-1)(x^2-3x+2) + \cos x & x < 0 \\ (x^2-1)(x^2-3x+2) + \cos x & 0 \le x < 1 \\ -(x^2-1)(x^2-3x+2) + \cos x & 1 \le x < 2 \\ (x^2-1)(x^2-3x+2) + \cos x & 2 \le x \end{cases}$$

$$f(x) = \begin{cases} (x^4-3x^3+x^2+3x-2) + \cos x & x < 1 \\ -(x^4-3x^3+x^2+3x-2) + \cos x & 1 \le x < 2 \\ (x^4-3x^3+x^2+3x-2) + \cos x & 2 \le x \end{cases}$$

$$f'(x) = \begin{cases} (4x^3-9x^2+2x+3) - \sin x & x < 1 \\ -(4x^3-9x^2+2x+3) - \sin x & 1 \le x < 2 \\ (4x^3-9x^2+2x+3) - \sin x & 2 \le x \end{cases}$$

At $x = 1$ At $x = 2$

$f'(1^-) = 0 = f'(1^+)$ $\qquad$ $f'(2^-) = -3$ & $f'(2^+) = 3$

Not differentiable

10. If $f(x) = \begin{cases} x\,e^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x \neq 0 \\ 0 \quad\quad\quad , & x = 0 \end{cases}$ then f(x) is :

(A) discontinuous everywhere

(B) continuous as well as differentiable for all x

(C) continuous for all x but not differentiable at $x = 0$

(D) neither differentiable nor continuous at $x = 0$

यदि $f(x) = \begin{cases} x\,e^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x \neq 0 \\ 0 \quad\quad\quad , & x = 0 \end{cases}$ तो f(x) है :

(A) सभी जगह असंतत्

(B) सभी x के लिऐ संतत् तथा अवकलनीय

(C) सभी x के लिऐ संतत परन्तु $x = 0$ पर अवकलनीय नहीं

(D) $x = 0$ पर ना तो अवकलनीय और ना ही संतत्

Ans. C

Sol. $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)}, & x \neq 0 \\ 0 \quad\quad , & x = 0 \end{cases}$ $\qquad \therefore\ |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

so $f(x) = \begin{cases} xe^{-2/x}, & x > 0 \\ x \quad\quad , & x < 0 \\ 0 \quad\quad , & x = 0 \end{cases}$

(I) continuous at $x = 0$

$\lim\limits_{h\to 0} f(0+h) = \lim\limits_{h\to 0} f(0-h) = f(0)$

$\lim\limits_{h\to 0} f(0+h) = \lim\limits_{h\to 0} h \times e^{-2/h} = 0$

$\lim\limits_{h\to 0} f(0-h) = \lim\limits_{h\to 0} = 0$ $\qquad$ $f(0) = 0$

f(x) is continuous at $x = 0$ or f(x) is continuous for all x

(II) differentiability at $x = 0$

$$\text{L.H.D.} = Lf'(0) = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h} = \frac{-h-0}{-h} = 1$$

$$\text{R.H.D. } Rf'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h}$$

$\Rightarrow \quad \dfrac{h \times e^{-2/h} - 0}{h} = e^{-2/h} = 0$

$Lf'(0) \neq Rf'(0)$

$f(x)$ is not differentiable at $x = 0$

so that $f(x)$ is cont at $x = 0$ but not differentiable at $x = 0$

11. $f(x) = \begin{cases} -x & x < 0 \\ x^2 & 0 \le x \le 1 \\ x^3 - x^2 + 1 & x > 1 \end{cases}$ examine differentiability of $f(x)$ at $x = 0, 1$.

$f(x) = \begin{cases} -x & x < 0 \\ x^2 & 0 \le x \le 1 \\ x^3 - x^2 + 1 & x > 1 \end{cases}$ तो $f(x)$ की $x = 0, 1$ पर अवकलनीयता की जाँच कीजिये।

Ans. non diff at $x = 0$, non diff at $x = 1$

Sol. $f(x) = \begin{cases} -x & x < 0 \\ x^2 & 0 \le x \le 1 \\ x^3 - x^2 + 1 & x > 1 \end{cases}$ differentiable at 0,1

$f'(x) = \begin{cases} -1 & x < 0 \\ 2x & 0 \le x \le 1 \\ 3x^2 - 2x & x > 1 \end{cases}$

At $x = 0$	at $x = 1$
$f'(0^-) = -1$	$f'(1^-) = 2$
$f'(0^+) = 0$	$f'(1^+) = 1$
LHD $\neq$ RHD but finite,	$2 \neq 1$ But finite
continuous but not differentiable	not differentiable but continuous

12. A function 'f' is defined as $f(x) = \begin{cases} \dfrac{1}{|x|} & \text{if} \quad |x| \ge \frac{1}{2} \\ ax^2 + bx + c & \text{if} \quad |x| < \frac{1}{2} \end{cases}$. If 'f' is differentiable at $x = \dfrac{1}{2}$ and $x = -\dfrac{1}{2}$,

then find the values of a, b and c if possible.

यदि $f(x) = \begin{cases} \frac{1}{|x|} & \text{if} \quad |x| \geq \frac{1}{2} \\ ax^2 + bx + c & \text{if} \quad |x| < \frac{1}{2} \end{cases}$. यदि 'f', $x = \frac{1}{2}$ तथा $x = -\frac{1}{2}$ पर अवकलनीय है, तो a, b तथा c के सम्भव मान होंगे।

Ans. $a = -4; \quad b = 0; \quad c = 3$

Sol. $f(x) = \begin{cases} \frac{1}{|x|} & |x| \geq \frac{1}{2} \\ ax^2 + bx + c & |x| < \frac{1}{2} \end{cases} = \begin{cases} -\frac{1}{x} & x \leq \frac{1}{2} \\ ax^2 + bx + c & -\frac{1}{2} < x < \frac{1}{2} \\ \frac{1}{x} & \frac{1}{2} \leq x \end{cases}$

differentiable at $x = -\frac{1}{2}$ differentiable at $x = \frac{1}{2}$

$f'(x) = \begin{cases} \frac{1}{x^2} & x \leq \frac{1}{2} \\ 2ax + b & -\frac{1}{2} < x < \frac{1}{2} \\ -\frac{1}{x^2} & \frac{1}{2} \leq x \end{cases}$

differentiable at $x = -\frac{1}{2}$ differentiable at $x = \frac{1}{2}$

$f'\left(-\frac{1}{2}^-\right) = f'\left(-\frac{1}{2}^+\right)$ $f'\left(\frac{1}{2}^-\right) = f'\left(\frac{1}{2}^+\right)$

$4 = -a + b$ $2a\left(\frac{1}{2}\right) + b = -4$

$a - b = -4$ ...(1) $a + b = -4$ ...(2)

(1) + (2); $a = -4, b = 0$

f(x) is continuous at $x = \frac{1}{2}$

$\frac{a}{4}+\frac{b}{2}+c=2$

$-1+c=2$

$c=3$

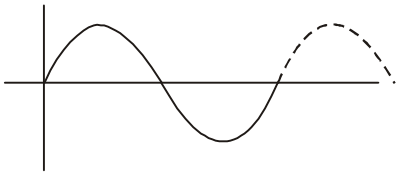
13. Discuss the continuity and differentiability of the function $f(x)=\sin x+\sin|x|$, $x\in R$. Draw a rough sketch of the graph of f(x).

फलन $f(x)=\sin x+\sin|x|$, $x\in R$ की सतत्ता तथा अवकलनीयता की जाँच करो। फलन f (x) का कच्चा रेखाचित्र भी खीचे।

Ans. f(x) is conti. but not derivable at x = 0

Sol. $f(x)=\sin x+\sin|x|=\begin{cases}\sin x-\sin x=0 & x<0\\ 2\sin x & x\geq 0\end{cases}$

continuous at x = 0 but not differentiable



14. Examine the continuity and differentiability of $f(x)=|x|+|x-1|+|x-2|$ $x\in R$. Also draw the graph of f(x).

फलन $f(x)=|x|+|x-1|+|x-2|$ $x\in R$ की सतत्ता तथा अवकलनीयता की जाँच करो। एवं f(x) का रेखाचित्र भी बनाओं।

Ans. cont. $\forall$ $x\in R$, not diff. at x = 0,1 and 2

Sol. $f(x)=|x|+|x-1|+|x-2|$

$$f(x)=\begin{cases}-x-(x-1)-(x-2)=-3x+3 & x<0\\ x-(x-1)-(x-2)=3-x & 0\leq x<1\\ x+(x-1)-(x-2)=x+1 & 1\leq x<2\\ x+(x-1)+(x-2)=3x-3 & 2\leq x\end{cases}$$

continuity

At x = 0	At x = 1	At x = 2
$f(0^-)=3$	$f(1^-)=2$	$f(2^-)=3$
$f(0^+)=3$	$f(1^+)=2$	$f(2^+)=3$
$f(0)=3$	$f(1)=2$	$f(2)=3$

f(x) is continuous but not differentiable at x = 0,1,2 (sharp edges)

1. The number of point where function f(x) = minimum $\{x^3 - 1, -x + 1, \text{sgn}(-x)\}$ is continuous but not differentiable is :

(A) One (B) Two (C) Zero (D) None of these

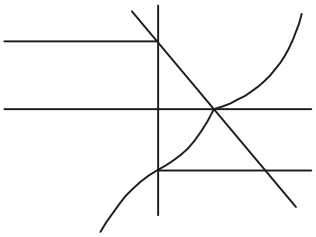
उन बिन्दुओं की संख्या जहाँ फलन f(x) = minimum $\{x^3 - 1, -x + 1, \text{sgn}(-x)\}$ संतत है परन्तु अवकलनीय नहीं :

(A) 1 (B) 2 (C) शून्य (D) इनमें से कोई नहीं

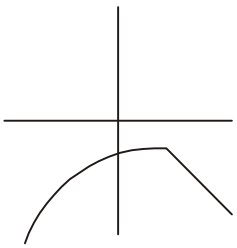
Ans. A

Sol. $f(x) = \min\{x^3 - 1, -x + 1, \text{sgn}(-x)\}$

$$f(x) = \begin{cases} x^3 - 1 & x < 0 \\ -1 & x = 0 \\ -1 & 0 < x < 2 \\ -x + 1 & 2 \le x \end{cases}$$



$$f(x) = \begin{cases} x^3 - 1 & x < 0 \\ -1 & 0 \le x < 2 \\ -x + 1 & 2 \le x \end{cases}$$



Not differentiable at x = 2

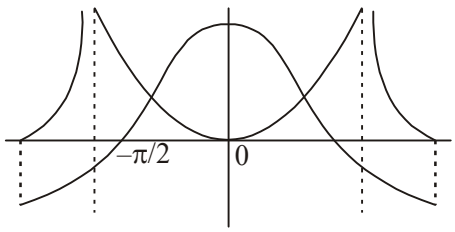
2. Number of points where the function f(x) = max (|tan x|, cos |x|) is non differentiable in the interval $(-\pi, \pi)$ is :

उन बिन्दुओं की संख्या जहाँ पर फलन f(x) = max (|tan x|, cos |x|) अन्तराल $(-\pi, \pi)$ में अवकलनीय नहीं है :

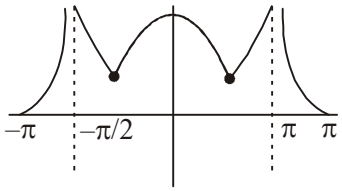
(A) 4 (B) 6 (C) 3 (D) 2

Ans. A

Sol.



$f(x) = \max \{|\tan x|, \cos |x|\}$



not differentiable at 4 point

3. If both f(x) & g(x) are differentiable functions at $x = x_0$, then the function defined as, h(x) = Maximum {f(x), g(x)} :

(A) is always differentiable at $x = x_0$

(B) is never differentiable at $x = x_0$

(C) is differentiable at $x = x_0$ if $f(x_0) \neq g(x_0)$

(D) cannot be differentiable at $x = x_0$ if $f(x_0) = g(x_0)$

यदि f(x) तथा g(x), $x = x_0$ पर अवकलनीय है तथा यदि फलन h(x) = Maximum {f(x), g(x)} तो यह होगा :

(A) $x = x_0$ पर हमेशा अवकलनीय है :

(B) $x = x_0$ पर कभी भी अवकलनीय नहीं है

(C) $x = x_0$ पर अवकलनीय है यदि $f(x_0) \neq g(x_0)$

(D) $x = x_0$ पर अवकलनीय नहीं होगा यदि $f(x_0) = g(x_0)$.

Ans. C

Sol. Consider the graph of $h(x) = \max(x, x^2)$ at $x = 0$ and $x = 1$
for D: $h(x) = \max. (x^2, -x^2)$]

4. Suppose that f is a differentiable function with the property that $f(x + y) = f(x) + f(y) + xy$ and $\lim\limits_{h\to 0} \frac{1}{h} f(h) = 3$ then:

(A) f is linear function (B) $f(x) = 3x + x^2$ (C) $f(x) = 3x + x^2/2$ (D) none of these

यदि f एक अवकलनीय फलन है जिसके लिये $f(x + y) = f(x) + f(y) + xy$ तथा $\lim\limits_{h\to 0} \frac{1}{h} f(h) = 3$ तो :

(A) f एक रेखीय फलन है (B) $f(x) = 3x + x^2$ (C) $f(x) = 3x + x^2/2$ (D) कोई नहीं

Ans. C

Sol. $f(x + y) = f(x) + f(y) + xy \Rightarrow f(0) = 0$

$$\lim_{h\to 0} \frac{f(h)}{h} = 3 \Rightarrow f'(0) = 3$$

$$\lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x)+f(h)+xh-f(x)}{h}$$

$$f'(x) = \lim_{h\to 0}\left(\frac{f(h)}{h}+x\right) = 3+x$$

$$f(x) = 3x + \frac{x^2}{2} + C$$

$$f(0) = 0 \Rightarrow f(x) = \frac{x^2}{2} + 3x$$

5. Let f: [0, 1] $\to$ [0, 1] be a continuous function. Then :
(A) f (x) = x for atleast one $0 \le x \le 1$
(B) f (x) will be differentiable in [0, 1]
(C) f (x) + x = 0 for atleast one x such that $0 \le x \le 1$
(D) none of these

माना f: [0, 1] $\to$ [0, 1] एक संत्त फलन है तो :
(A) f (x) = x होगा, कम से कम x के एक मान के लिए जहाँ $0 \le x \le 1$
(B) f (x), [0, 1] में अवकलनीय नहीं है
(C) f (x) + x = 0 होगा, कम से कम एक x के लिए जहाँ $0 \le x \le 1$
(D) इनमें में से कोई नहीं

Ans. A

Sol. $f : [0,1] \to [0,1]$

f(x) = continuous function

6. If f′(3) = 2 find the value of $\lim_{h\to 0} \frac{f(3+h^2)-f(3-h^2)}{2h^2}$.

यदि f′(3) = 2 तो $\lim_{h\to 0} \frac{f(3+h^2)-f(3-h^2)}{2h^2}$ का मान होगा।

Ans. 2

Sol. $f'(3) = 2$

$$= \lim_{h\to 0} \frac{f\left(3+h^2\right)-f\left(3-h^2\right)}{2h^2} = \lim_{h\to 0} \frac{f(3+h)-f(3-h)}{h}$$

$$= \lim_{h\to 0} f'(3+h) + f'(3-h) = 2f'(3) = 2\times 2 = 4$$

7. Check the differentiability of $f(x) = \begin{cases} (x-e)2^{-2^{\left(\frac{1}{e-x}\right)}} & , \quad x \neq e \\ 0 & , \quad x = e \end{cases}$ at $x = e$.

फलन $f(x) = \begin{cases} (x-e)2^{-2^{\left(\frac{1}{e-x}\right)}} & , \quad x \neq e \\ 0 & , \quad x = e \end{cases}$ की x = e पर अवकलनीयता की जाँच कीजिये।

Ans. non-diff.

Sol. $f(x) = \begin{cases} (x-e)2^{-2^{\left(\frac{1}{e-x}\right)}} & x \neq e \\ 0 & x = e \end{cases}$

At $x = e$

$$\text{LHD} = \lim_{h\to 0} \frac{(e-h-e)2^{-2^{\left(\frac{1}{e-(e-h)}\right)}} - 0}{-h}$$

$$= \lim_{h\to 0} \frac{-h\cdot 2^{-2^{\left(\frac{1}{h}\right)}}}{-h} = \lim_{h\to 0} 2^{-2^{\left(\frac{1}{h}\right)}} = 0$$

$$\text{RHD} = \lim_{h\to 0} \frac{h2^{-2^{-\frac{1}{h}}}}{h} = \lim_{h\to 0} \frac{1}{2^{2^{-\frac{1}{h}}}} = 1$$

Not differentiable

8. Let $f(x) = \frac{xg(x)}{|x|}$, $g(0) = g'(0) = 0$ and f(x) be continuous at x = 0. Find f'(0), if it exists.

यदि $f(x) = \frac{xg(x)}{|x|}$, $g(0) = g'(0) = 0$ तथा f(x), x = 0 संत्त है तो f'(0) का मान क्या होगा यदि यह विद्यमान है।

Ans. 0

Sol. $f(x)=\dfrac{xg(x)}{|x|}$ $\qquad g(0)=g'(0)=0$

f(x) is continuous at x = 0, f'(0) = ?

$$f'(0)=\lim_{h\to 0}\frac{f(h)-f(0)}{h}=\lim_{h\to 0}\frac{hg(h)}{|h|}=0$$

9. Let $f(x)=\begin{cases}|1-4x^2|, & 0\le x<1\\ [x^2-2x], & 1\le x<2\end{cases}$ where [.] denotes the greatest integer function . Then discuss the differentiability of $f(x)$ in $(0, 2)$.

यदि $f(x)=\begin{cases}|1-4x^2|, & 0\le x<1\\ [x^2-2x], & 1\le x<2\end{cases}$ (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) तो $f(x)$ की अन्तराल $(0, 2)$ में अवकलनीयता की जाँच कीजिये।

Ans. Non derivable at x = 1/2 and x = 1

Sol. $f(x)=\begin{cases}|1-4x^2| & 0\le x<1\\ [x^2-2x] & 1\le x<2\end{cases}$

At x = 1

$$\text{LHL}=\lim_{h\to 0}f(1-h)=\lim_{h\to 0}|1-4(1-h)^2|=3$$

$$\text{RHL}=\lim_{h\to 0}f(1+h)=\lim_{h\to 0}[(1+h)^2-2(1+h)]=-1$$

LHL ≠ RHL ⇒ Not continuous at x = 1 ⇒ not derivable at x = 1

$1-4x^2=0 \Rightarrow x=\pm\dfrac{1}{2} \Rightarrow$ at $x=\dfrac{1}{2}$, f(x) is not differentiable

$g(x)=x^2-2x$	min at x = 1	max at x = 2
(1, –1) Vertex	g(1) = –1	g(2) = 0
	$g(x)\in[-1,0)$	$\forall\ x\in[1,2)$
	$[g(x)]=-1$	$\forall\ x\in[1,2)$

10. Let $f(x)$ be defined in the interval $[-2, 2]$ such that, $f(x)=\begin{cases}[x] & -2 \le x \le -1/2 \\ 2x^2-1 & -1/2 < x \le 2\end{cases}$ and $g(x)=f(|x|)+|f(x)|$ where $[.]$ denotes the greatest integer function. Test for differentiability of $g(x)$ in $(-2, 2)$.

यदि $f(x)$ अन्तराल $[-2, 2]$ में फलन इस प्रकार परिभाषित है $f(x)=\begin{cases}[x] & -2 \le x \le -1/2 \\ 2x^2-1 & -1/2 < x \le 2\end{cases}$ तथा $g(x)=f(|x|+|f(x)|$ (जहाँ $[.]$ महत्तम पूर्णांक फलन दर्शाता है) तो $g(x)$ का $(-2, 2)$ अन्तराल में अवकलनीयता की जाँच करो।

Ans. $g(x)$ is not diff. at $x=-1,-\frac{1}{2},\frac{1}{\sqrt{2}}$

Sol. $f(x)=\begin{cases}[x] & -2 \le x \le -\frac{1}{2} \\ 2x^2-1 & -\frac{1}{2} < x \le 2\end{cases}$ $\qquad g(x)=f(|x|+|f(x)|$

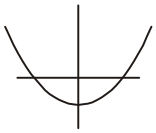
$$f(x)=\begin{cases}-2 & -2 \le x < -1 \\ -1 & -1 \le x \le -\frac{1}{2} \\ 2x^2-1 & -\frac{1}{2} < x \le 2\end{cases}$$

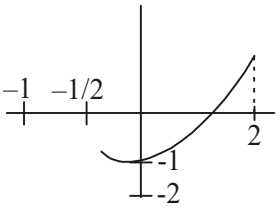
Discontinuous at $x=-1$, $x=-\frac{1}{2}$

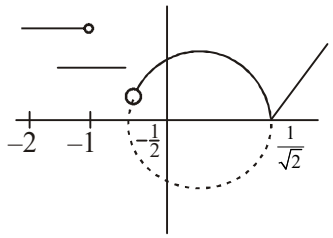
$$g(x)=\begin{cases}2+2x^2-1 & -2 \le x < -1 \\ 1+2x^2-1 & -1 \le x \le -\frac{1}{2} \\ |2x^2-1|+2x^2-1 & -\frac{1}{2} < x \le 2\end{cases}$$

$0 \qquad x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$2(2x^2-1) \qquad x \in \left(-\frac{1}{2}, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 2\right]$

f(|x|) = 

f(x)= 

|f(x)| = 

Discontinuous at $-1,-\frac{1}{2},\frac{1}{\sqrt{2}}$

11. $f(x)=\begin{cases} a+(x-b)^2 & , \ |x-b|\le k \\ c+|x-b| & , \ |x-b|>k \end{cases}$ find the value of 'k'. So that f (x) becomes differentiable at x = b – k, b + k and prove that a – c = 1/4.

$f(x)=\begin{cases} a+(x-b)^2 & , \ |x-b|\le k \\ c+|x-b| & , \ |x-b|>k \end{cases}$ में 'k' का मान क्या होगा। यदि फलन f (x), x = b – k , b + k पर अवकलनीय है तथा सिद्ध कीजिये a – c = 1/4

Ans. $\frac{1}{2}$

Sol. $f(x)=\begin{cases} a+(x-b)^2 & |x-b|\le k \\ c+|x-b| & |x-b|>k \end{cases}$

$f(x)=\begin{cases} a+(x-b)^2 & b-k<x<b+k \\ c+|x-b| & x-b\in(-\infty,-k)\cup(k,\infty) \end{cases}$

$$\Rightarrow x \in (-\infty, b-k) \cup (b+k, \infty)$$

$$f(x) = \begin{cases} a + (x-b)^2 & b-k < x < b+k \\ c+b-x & x \in (-\infty, b-k) \\ c+x-b & x \in (b+k, \infty) \end{cases}$$

At $x = b-k$

$$LHL = \lim_{h \to 0} (c+b) - (b-k+h) = c+k$$

$$RHL = \lim_{h \to 0} a + (b-k-b)^2 = a+k^2$$

$k^2 + a = c + k \Rightarrow a - c = k - k^2$

At $x = b+k$

$$LHL = \lim_{h \to 0} a + (b+k-h-b)^2 = a+k^2$$

$$RHL = \lim_{h \to 0} c + (b+k-h) - b = c+k$$

$$f'(x) = \begin{cases} 2(x-b) & b-k < x < b+k \\ -1 & x \in (-\infty, b-k) \\ 1 & x \in (b+k, \infty) \end{cases}$$

$f'\left((b+k)^-\right) = f'\left((b+k)^+\right)$

$2\ (b + k - b) = +1$

$2k = 1 \Rightarrow k = \frac{1}{2}$

$f'\left((b-k)^-\right) = f'\left((b-k)^+\right)$

$2(b - k - b) = -1$

$2k = 1$

$k = \frac{1}{2}$

$a - c = k - k^2$

$a - c = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$

$$a - c = \frac{1}{4}$$

12. Let $f(x) = x^3 - 9x^2 + 15x + 6$ and $g(x) = \begin{cases} \text{Min. } \{ f(t) \ ; \ 0 \le t \le x \} & , \ 0 \le x \le 6 \\ x - 18 & , \ x > 6 \end{cases}$. Then draw the graph of $g(x)$ and discuss its continuity and differentiability.

यदि $f(x) = x^3 - 9x^2 + 15x + 6$ तथा $g(x) = \begin{cases} \text{Min. } \{ f(t) \ ; \ 0 \le t \le x \} & , \ 0 \le x \le 6 \\ x - 18 & , \ x > 6 \end{cases}$, तो $g(x)$ का ग्राफ बनाकर इसकी अवकलनीयता तथा सांत्यतता की जाँच करें।

Ans.

Sol. $f(x) = x^3 - 9x^2 + 15x + 6$

$$g(x) = \begin{cases} \text{Min. } \{ f(t) \ ; \ 0 \le t \le x \} & , \ 0 \le x \le 6 \\ x - 18 & , \ x > 6 \end{cases}$$

$f'(x) = 3x^2 - 18x + 15 \rightarrow$ root exist

$f(x) \Rightarrow$ Three zeroes $\qquad f(x) = x^3 - 9x^2 + 15x + 6$

$f(0) = 6, f(3) = -3$

$f(6) = -12, f(-1) = -19$

one root between (0, 3) & one root between (–1,0)

$f'(x) = 3x^2 - 18x + 15$

$3x^2 - 3x - 15x + 15 = 0$

$3x(x - 1) - 15(x - 1) = 0$

$(3x - 15)(x - 1) = 0$

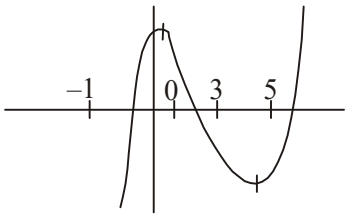
$x = 5, x = 1$

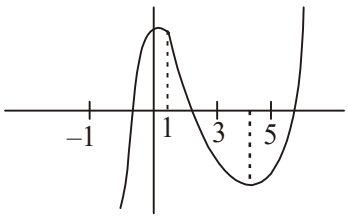
min $\{f(t) : 0 \le t < x\}$

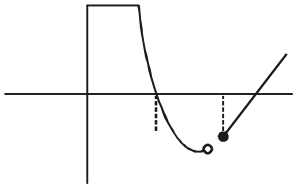
$$\begin{cases} f(0) = 6 & 0 \le t < 1 \\ f(t) & 1 \le t < 5 \\ f(t) = -19 & 5 \le t < 6 \end{cases}$$

Not differential at $x = 1$

discontinuous at $x = 6$







13. Examine for continuity & differentiability the points x = 1 and x = 2, the function f defined by $f(x)=\begin{cases} x[x] & , \ 0\le x<2 \\ (x-1)[x] & , \ 2\le x\le 3 \end{cases}$ where [x] = greatest integer less than or equal to x.

फलन f(x) की x = 1 तथा x = 2 पर सतत्ता तथा अवकलनीयता की विवेचना करो यहा पर $f(x)=\begin{cases} x[x] & , \ 0\le x<2 \\ (x-1)[x] & , \ 2\le x\le 3 \end{cases}$ जहाँ [x] महत्तम पूर्णांक फलन है :

Ans. discontinuous and not derivable at x = 1, continuous but not derivable at x = 2

Sol. x = 1, x = 2

$$f(x)=\begin{cases} x[x] & 0\le x<2 \\ (x-1)[x] & 0\le x\le 3 \end{cases}$$

$$f(x)=\begin{cases} 0 & 0\le x<1 \\ x & 1\le x<2 \\ 2(x-1) & 2\le x<3 \\ 6 & x=3 \end{cases}$$

discontinuous at x = 1,3, continuous at x = 2

$$f'(x)=\begin{cases} 0 & 0\le x<1 \\ 1 & 1\le x<2 \\ 2 & 2\le x<3 \end{cases}$$

not differentiable at x = 2